PYTHON FUNDAMENTALS

Answer Key for the 200 Practice Questions

Concise reference solutions, numbered to match the companion workbook.

How to use this answer key

Try each question independently before checking the solution. These are example solutions, not the only correct answers. Some examples use sample filenames, values, or grading thresholds; adapt them to the exact requirements of your version of the problem.

Run code snippets in a Python 3 environment. Where a solution is shown as a fragment or comment, it demonstrates the core idea and may need to be placed into a complete script. Be especially careful with file paths and destructive operations such as deleting files.

Tip: Don't copy-paste your way through the workbook. First attempt the problem, then compare your approach with the reference solution and test edge cases.

Input, output, variables, and arithmetic

Question 1

```
name = input("Name: ")
age = int(input("Age: "))
print(f"Hello, {name}! You are {age}.")
```

Question 2

```
a, b = float(input("First: ")), float(input("Second: "))
print("sum", a+b, "difference", a-b, "product", a*b)
if b != 0: print("quotient", a/b, "remainder", a%b)
else: print("Cannot divide by zero.")
```

Question 3

```
c = float(input("Celsius: "))
print(c * 9 / 5 + 32)
```

Question 4

```
l, w = float(input("Length: ")), float(input("Width: "))
print("Area:", l*w, "Perimeter:", 2*(l+w))
```

Question 5

```
import math
r = float(input("Radius: "))
print("Area:", math.pi*r*r, "Circumference:", 2*math.pi*r)
```

Question 6

```
t = int(input("Seconds: "))
print(t//3600, "hours", (t%3600)//60, "minutes", t%60, "seconds")
```

Question 7

```
a, b = 3, 8
a, b = b, a
print(a, b)  # Python tuple unpacking swaps values
```

Question 8

```
total = float(input("Bill: "))
pct = float(input("Tip percent: "))
tip = total*pct/100
print(f"Tip: {tip:.2f}; total: {total+tip:.2f}")
```

Question 9

```
days = int(input("Days: "))
print(days//7, "weeks and", days%7, "days")
```

Question 10

```
n = abs(int(input("Three-digit integer: ")))
print(n//100 + (n//10)%10 + n%10)
```

Conditions and Boolean logic

Question 11

```
n = int(input("Integer: "))
print("positive" if n>0 else "negative" if n<0 else "zero")
```

Question 12

```
n = int(input("Integer: "))
print("even" if n%2==0 else "odd")
```

Question 13

```
a, b = map(float, input("Two numbers: ").split())
print("equal" if a==b else max(a,b))
```

Question 14

```
a,b,c = map(float,input("Three numbers: ").split())
largest = a
if b>largest: largest=b
if c>largest: largest=c
print(largest)
```

Question 15

```
y = int(input("Year: "))
print(y%400==0 or (y%4==0 and y%100!=0))
```

Question 16

```
ch = input("Character: ").lower()
print("vowel" if ch in "aeiou" and len(ch)==1 else "consonant" if ch.isalpha() and len(ch)==1 else "not
one letter")
```

Question 17

```
m = float(input("Marks: "))
print("A" if m>=90 else "B" if m>=80 else "C" if m>=70 else "D" if m>=60 else "F")
```

Question 18

```
n = int(input("Number: "))
print(n%5==0 and n%11==0)
```

Question 19

```
a,b,c = map(float,input("Three sides: ").split())
valid = min(a,b,c)>0 and a+b>c and a+c>b and b+c>a
print("Valid triangle" if valid else "Invalid triangle")
```

Question 20

```
a,b,c = sorted(map(float,input("Three sides: ").split()))
if a>0 and a+b>c: print("equilateral" if a==c else "isosceles" if a==b or b==c else "scalene")
else: print("invalid triangle")
```

Question 21

```
units = float(input("Units used: "))
if units<=100: bill=units*1.5
elif units<=300: bill=150+(units-100)*2.5
else: bill=650+(units-300)*4
print(f"Bill: {bill:.2f}")
```

Question 22

```
user = input("Username: "); password = input("Password: ")
print("Match" if user=="admin" and password=="python123" else "No match")
```

Question 23

```
ch = input("Character: ")
print("upper" if ch.isupper() else "lower" if ch.islower() else "digit" if ch.isdigit() else "special/
other")
```

Question 24

```
n = float(input("Number: ")); low, high = 10, 50
print(low <= n <= high)
```

Question 25

```
a,b = map(float,input("Two numbers: ").split()); op=input("Operator (+,-,*,/): ")
if op=="+": print(a+b)
elif op=="-": print(a-b)
elif op=="*": print(a*b)
elif op=="/" and b!=0: print(a/b)
else: print("Invalid operator or division by zero")
```

Loops, number logic, and series

Question 26

```
for n in range(1, 101): print(n)
```

Question 27

```
for n in range(100, 0, -1): print(n)
```

Question 28

```
n=int(input("n: ")); print(sum(range(1,n+1)))
```

Question 29

```
n=int(input("n: ")); result=1
for i in range(2,n+1): result*=i
print(result)
```

Question 30

```
n=int(input("Number: "))
for i in range(1,11): print(f"{n} x {i} = {n*i}")
```

Question 31

```
a,b=map(int,input("Start end: ").split())
print([x for x in range(a,b+1) if x%2==0])
```

Question 32

```
a,b=map(int,input("Start end: ").split())
print([x for x in range(a,b+1) if x%2!=0])
```

Question 33

```
n=abs(int(input("Integer: "))); count=1 if n==0 else 0
while n: count+=1; n//=10
print(count)
```

Question 34

```
n=abs(int(input("Integer: "))); rev=0
while n: rev=rev*10+n%10; n//=10
print(rev)
```

Question 35

```
n=input("Integer: ").strip(); print(n==n[::-1])
```

Question 36

```
n=abs(int(input("Integer: "))); total=0
while n: total+=n%10; n//=10
print(total)
```

Question 37

```
n=abs(int(input("Integer: "))); product=1
if n==0: product=0
while n: product*=n%10; n//=10
print(product)
```

Question 38

```
n=int(input("Number: ")); prime=n>=2
for d in range(2,int(n**0.5)+1):
 if n%d==0: prime=False; break
print(prime)
```

Question 39

```
a,b=map(int,input("Range: ").split())
for n in range(max(2,a),b+1):
 if all(n%d for d in range(2,int(n**0.5)+1)): print(n)
```

Question 40

```
a,b=map(int,input("Two positive integers: ").split())
while b: a,b=b,a%b
print(a)
```

Question 41

```
a,b=map(int,input("Two positive integers: ").split())
x,y=a,b
while y: x,y=y,x%y
print(abs(a*b)//x if x else 0)
```

Question 42

```
n=int(input("Terms: ")); a,b=0,1
for _ in range(n): print(a,end=" "); a,b=b,a+b
print()
```

Question 43

```
n=int(input("Number: ")); a,b=0,1
while a<n: a,b=b,a+b
print(n==a or n==0)
```

Question 44

```
n=int(input("Number: ")); s=str(abs(n)); print(sum(int(d)**len(s) for d in s)==n)
```

Question 45

```
a,b=map(int,input("Range: ").split())
for n in range(a,b+1):
 s=str(n)
 if n>=0 and sum(int(d)**len(s) for d in s)==n: print(n)
```

Question 46

```
n=int(input("Number: ")); print(n>0 and sum(d for d in range(1,n) if n%d==0)==n)
```

Question 47

```
a,b=map(int,input("Range: ").split())
for n in range(max(2,a),b+1):
 if sum(d for d in range(1,n) if n%d==0)==n: print(n)
```

Question 48

```
n=int(input("Terms: ")); print(sum(1/i for i in range(1,n+1)))
```

Question 49

```
n=int(input("n: ")); loop_sum=0
for i in range(1,n+1): loop_sum+=i
print(loop_sum, n*(n+1)//2)
```

Question 50

```
count=total=0
while True:
 s=input("Number (q to stop): ")
 if s.lower()=="q": break
 x=float(s); count+=1; total+=x
print("count",count,"total",total,"average",total/count if count else 0)
```

Nested loops and patterns

Question 51

```
n=int(input("Side: "))
for _ in range(n): print("* "*n)
```

Question 52

```
n=int(input("Height: "))
for i in range(1,n+1): print("*"*i)
```

Question 53

```
n=int(input("Height: "))
for i in range(n,0,-1): print("*"*i)
```

Question 54

```
n=int(input("Height: "))
for i in range(1,n+1): print(" "*(n-i)+"*"*(2*i-1))
```

Question 55

```
n=int(input("Height: "))
for i in range(n,0,-1): print(" "*(n-i)+"*"*(2*i-1))
```

Question 56

```
n=int(input("Side: "))
for i in range(n): print("*"*n if i in (0,n-1) else "*"+" "*(n-2)+"*")
```

Question 57

```
n=int(input("Rows: "))
for i in range(1,n+1): print(" ".join(str(x) for x in range(1,i+1)))
```

Question 58

```
n=int(input("Rows: "))
for i in range(1,n+1): print(" ".join([str(i)]*i))
```

Question 59

```
n=int(input("Rows: ")); value=1
for i in range(1,n+1):
 for _ in range(i): print(value,end=" "); value+=1
 print()
```

Question 60

```
n=int(input("Half-height: "))
for i in list(range(1,n+1))+list(range(n-1,0,-1)): print(" "*(n-i)+"*"*(2*i-1))
```

Question 61

```
for i in range(1,11): print(" ".join(f"{i*j:4}" for j in range(1,11)))
```

Question 62

```
n=int(input("Size: "))
for i in range(n): print("".join("#" if (i+j)%2==0 else "." for j in range(n)))
```

Lists and list operations

Question 63

```
nums=list(map(float,input("Numbers separated by spaces: ").split()))
print(nums, len(nums), sum(nums))
```

Question 64

```
nums=list(map(float,input("Numbers: ").split()))
if nums: print("largest",max(nums),"smallest",min(nums))
```

Question 65

```
nums=list(map(float,input("Numbers: ").split()))
print(sum(nums)/len(nums) if nums else "Empty list")
```

Question 66

```
nums=list(map(int,input("Numbers: ").split()))
print("even",sum(x%2==0 for x in nums),"odd",sum(x%2!=0 for x in nums))
```

Question 67

```
nums=list(map(int,input("Numbers: ").split())); print([x for x in nums if x>0])
```

Question 68

```
nums=list(map(int,input("Numbers: ").split())); unique=list(dict.fromkeys(nums)); print(unique)
```

Question 69

```
nums=list(map(int,input("Numbers: ").split())); out=[]
for x in nums: out.insert(0,x)
print(out)
```

Question 70

```
nums=list(map(int,input("Numbers: ").split())); target=int(input("Target: "))
print(nums.index(target) if target in nums else "Not found")
```

Question 71

```
nums=list(map(int,input("Numbers: ").split())); target=int(input("Value: ")); print(nums.count(target))
```

Question 72

```
nums=list(map(int,input("Numbers: ").split())); distinct=sorted(set(nums)); print(distinct[-2] if len(
distinct)>=2 else "No second-largest distinct value")
```

Question 73

```
a=list(map(int,input("First list: ").split())); b=list(map(int,input("Second list: ").split())); print(
sorted(a+b))
```

Question 74

```
nums=list(map(int,input("Numbers: ").split())); print([x for x in nums if x%2==0], [x for x in nums if
x%2!=0])
```

Question 75

```
nums=list(map(int,input("Numbers: ").split())); k=int(input("Rotate by: ")); k%=len(nums) if nums else 1;
 print(nums[-k:]+nums[:-k] if k else nums)
```

Question 76

```
nums=list(map(int,input("Numbers: ").split())); print([x for x in nums if x!=0]+[x for x in nums if
x==0])
```

Question 77

```
a=list(map(int,input("First list: ").split())); b=list(map(int,input("Second list: ").split())); print(
list(dict.fromkeys(x for x in a if x in b)))
```

Question 78

```
a=list(map(int,input("First list: ").split())); b=set(map(int,input("Second list: ").split())); print([x
for x in dict.fromkeys(a) if x not in b])
```

Question 79

```
nums=list(map(int,input("Numbers: ").split())); size=int(input("Chunk size: "))
print([nums[i:i+size] for i in range(0,len(nums),size)] if size>0 else "Size must be positive")
```

Question 80

```
nested=[[1,2],[3,4],[5]]; print([x for group in nested for x in group])
```

Question 81

```
nums=list(map(int,input("Numbers: ").split())); target=int(input("Target sum: "))
print([(nums[i],nums[j]) for i in range(len(nums)) for j in range(i+1,len(nums)) if nums[i]+nums[
j]==target])
```

Question 82

```
a=[[1],[2]]; alias=a; shallow=a.copy(); a[0].append(9); print("alias",alias,"shallow",shallow,"same
outer?",a is shallow)
```

Strings and text processing

Question 83

```
s=input("Sentence: "); print("with spaces",len(s),"without spaces",sum(not c.isspace() for c in s))
```

Question 84

```
s=input("Sentence: "); print(len(s.split()))
```

Question 85

```
s=input("Text: "); out=""
for ch in s: out=ch+out
print(out)
```

Question 86

```
import re
s=input("Text: "); cleaned=re.sub(r"[^a-z0-9]","",s.lower()); print(cleaned==cleaned[::-1])
```

Question 87

```
s=input("Text: ").lower(); print("vowels",sum(c in "aeiou" for c in s),"consonants",sum(c.isalpha() and
c not in "aeiou" for c in s))
```

Question 88

```
s=input("Text: "); print("upper",sum(c.isupper() for c in s),"lower",sum(c.islower() for c in s),
"digits",sum(c.isdigit() for c in s),"whitespace",sum(c.isspace() for c in s))
```

Question 89

```
from collections import Counter
print(Counter(input("Text: ")))
```

Question 90

```
from collections import Counter
s=input("Text: "); counts=Counter(s); print(next((c for c in s if counts[c]==1),"No non-repeating
character"))
```

Question 91

```
s=input("Text: "); print("".join(dict.fromkeys(s)))
```

Question 92

```
s=input("Text: "); old=input("Replace: "); new=input("With: "); print(s.replace(old,new))
```

Question 93

```
clean=lambda s: sorted(c.lower() for c in s if not c.isspace())
a=input("First: "); b=input("Second: "); print(clean(a)==clean(b))
```

Question 94

```
s=input("Sentence: "); words=s.split(); print(max(words,key=len) if words else "No words")
```

Question 95

```
s=input("Sentence: "); print(" ".join(w[:1].upper()+w[1:] for w in s.split()))
```

Question 96

```
s=input("Sentence: "); words=s.split(); sep=input("Separator: "); print(sep.join(words))
```

Question 97

```
import re
s=input("Sentence: "); word=input("Word: "); print(len(re.findall(r"\b"+re.escape(word)+r"\b",s,re.I)))
```

Question 98

```
s=input("Text: "); print(s.isdigit())
```

Question 99

```
s=input("Password: "); print(len(s)>=8 and any(c.islower() for c in s) and any(c.isupper() for c in s)
and any(c.isdigit() for c in s))
```

Question 100

```
import re
s=input("Text: "); print(re.findall(r"-?\d+",s))
```

Question 101

```
s=input("Text: "); out=""; i=0
while i<len(s):
 j=i+1
 while j<len(s) and s[j]==s[i]: j+=1
 out+=s[i]+str(j-i); i=j
print(out)
```

Question 102

```
name=input("Name: ").strip(); amount=float(input("Amount: ")); print(f"{name}: ${amount:,.2f}")
```

Tuples, sets, and dictionaries

Question 103

```
person=("Asha",20,"Pune"); name,age,city=person; print(name,age,city)
```

Question 104

```
def min_max(values):
 if not values: raise ValueError("Collection cannot be empty")
 return min(values),max(values)
print(min_max([4,1,9,2]))
```

Question 105

```
a=set(map(int,input("Set A: ").split())); b=set(map(int,input("Set B: ").split())); print(a|b,a&b,a-b,
a^b)
```

Question 106

```
nums=[3,1,3,2,1]; print(list(set(nums)))  # set order is not guaranteed
```

Question 107

```
a={1,2}; b={1,2,3}; print(a<=b, a.isdisjoint({4,5}))
```

Question 108

```
students={"Asha":88,"Ravi":76,"Mina":93}
for name,marks in students.items(): print(name,marks)
```

Question 109

```
d={"a":1}; d["b"]=2; d["a"]=3; print(d.get("missing","Not found"),d)
```

Question 110

```
s=input("Sentence: ").lower().split(); counts={}
for word in s: counts[word]=counts.get(word,0)+1
print(counts)
```

Question 111

```
book={}
while True:
 action=input("add/search/update/delete/quit: ").lower()
 if action=="quit": break
 name=input("Contact name: ")
 if action=="add" or action=="update": book[name]=input("Phone: ")
 elif action=="search": print(book.get(name,"Not found"))
 elif action=="delete": book.pop(name,None)
print(book)
```

Question 112

```
a={"x":1,"y":2}; b={"y":9,"z":3}; print(a|b)  # right-hand values win
```

Question 113

```
d={"a":4,"b":9,"c":2}; print(max(d,key=d.get) if d else None)
```

Question 114

```
d={"b":2,"a":3,"c":1}; print(sorted(d.items())); print(sorted(d.items(),key=lambda item:item[1]))
```

Question 115

```
words=["apple","ant","bat","berry"]; groups={}
for w in words: groups.setdefault(w[0],[]).append(w)
print(groups)
```

Question 116

```
keys=["a","b"]; values=[1,2]; print(dict(zip(keys,values)))  # zip stops at shorter list
```

Question 117

```
from collections import Counter
print(Counter(input("Text: ")))
```

Question 118

```
cart={"apples":3,"bread":2,"milk":1}; print("Total items:",sum(cart.values()))
```

Question 119

```
course1={"Asha","Ravi","Mina"}; course2={"Ravi","Mina","Zoya"}; print("both",course1&course2,"only one",(
course1|course2)-(course1&course2))
```

Question 120

```
students={"Asha":{"Math":90,"English":80},"Ravi":{"Math":70,"English":85}}
for name,marks in students.items(): print(name,sum(marks.values())/len(marks))
```

Functions, recursion, and scope

Question 121

```
def calculate(a,b): return a+b,a-b,a*b,(a/b if b else None)
print(calculate(8,2))
```

Question 122

```
def factorial(n):
 if n<0: raise ValueError("n must be non-negative")
 result=1
 for i in range(2,n+1): result*=i
 return result
print(factorial(5))
```

Question 123

```
def is_prime(n): return n>=2 and all(n%d for d in range(2,int(n**0.5)+1))
print(is_prime(29))
```

Question 124

```
def average(nums):
 if not nums: return None
 return sum(nums)/len(nums)
print(average([2,4,6]))
```

Question 125

```
def convert(value,unit="C"):
 return value*9/5+32 if unit.upper()=="C" else (value-32)*5/9
print(convert(25),convert(77,"F"))
```

Question 126

```
def greet(name,greeting="Hello"): return f"{greeting}, {name}!"
print(greet("Asha"),greet("Ravi","Welcome"))
```

Question 127

```
def add_all(*args): return sum(args)
print(add_all(1,2,3,4))
```

Question 128

```
def show(**kwargs):
 for key,value in kwargs.items(): print(key,value)
show(name="Asha",age=20)
```

Question 129

```
def stats(nums): return min(nums),max(nums),sum(nums)/len(nums)
lo,hi,avg=stats([2,4,6]); print(lo,hi,avg)
```

Question 130

```
def factorial(n): return 1 if n<=1 else n*factorial(n-1)
print(factorial(5))
```

Question 131

```
def fib(n): return n if n<2 else fib(n-1)+fib(n-2)
print(fib(8))  # simple recursive version; slow for large n
```

Question 132

```
def digit_sum(n): return n if n<10 else n%10+digit_sum(n//10)
print(digit_sum(abs(1234)))
```

Question 133

```
def reverse(s): return s if len(s)<2 else reverse(s[1:])+s[0]
print(reverse("python"))
```

Question 134

```
def binary_search(a,target,lo=0,hi=None):
 if hi is None: hi=len(a)-1
 if lo>hi: return -1
 mid=(lo+hi)//2
 if a[mid]==target: return mid
 return binary_search(a,target,lo,mid-1) if target<a[mid] else binary_search(a,target,mid+1,hi)
print(binary_search([1,3,5,7],5))
```

Question 135

```
def demo():
 local_value=10
 return local_value
print(demo())  # local_value exists only inside demo
```

Question 136

```
def apply_each(func,values): return [func(x) for x in values]
print(apply_each(abs,[-2,3,-4]))
```

Question 137

```
items=[("Asha",88),("Ravi",75),("Mina",93)]; print(sorted(items,key=lambda item:item[1]))
```

Question 138

```
def area(length: float, width: float) -> float:
    """Return rectangle area for non-negative dimensions."""
    if length<0 or width<0: raise ValueError("Dimensions must be non-negative")
    return length*width
print(area(3,4))
```

Core Python features and idioms

Question 139

```
n=int(input("Limit: ")); print([x*x for x in range(n+1) if x%2==0])
```

Question 140

```
n=int(input("Limit: ")); print({x:x*x for x in range(n+1)})
```

Question 141

```
s=input("Text: "); print({c for c in s.lower() if c in "aeiou"})
```

Question 142

```
items=["apple","banana","pear"]
for i,item in enumerate(items,start=1): print(i,item)
```

Question 143

```
a=[1,2,3]; b=["a","b"]; print(list(zip(a,b)))  # pairs only until shorter iterable ends
```

Question 144

```
words=input("Words: ").split(); print(sorted(words,key=lambda w:(len(w),w.lower())))
```

Question 145

```
nums=list(map(int,input("Numbers: ").split())); print("any positive?",any(x>0 for x in nums),"all
positive?",all(x>0 for x in nums))
```

Question 146

```
nums=list(map(int,input("Numbers: ").split())); print(list(map(lambda x:x*x,filter(lambda x:x%2==0,
nums))))
# Equivalent: [x*x for x in nums if x%2==0]
```

Question 147

```
def squares(n):
 for x in range(1,n+1): yield x*x
print(list(squares(5)))
```

Question 148

```
def evens(start,stop):
 for n in range(start,stop+1):
  if n%2==0: yield n
print(list(evens(1,10)))
```

Question 149

```
items=[10,20,30,40]; first,*middle,last=items; print(first,middle,last)
```

Question 150

```
def inspect_args(*args,**kwargs): print("positional:",args,"keywords:",kwargs)
inspect_args(1,2,name="Asha")
```

Question 151

```
try:
 value=int(input("Integer: "))
except ValueError: print("Please enter an integer.")
else: print("Value:",value)
finally: print("Finished.")
```

Question 152

```
def divide(a,b):
 if b==0: raise ValueError("b must not be zero")
 return a/b
try: print(divide(8,0))
except ValueError as err: print(err)
```

Question 153

```
x=4; assert x>0, "x must be positive"; print(x*x)
```

Question 154

```
import copy
a=[[1],[2]]; shallow=copy.copy(a); deep=copy.deepcopy(a); a[0].append(9); print(shallow,deep)
```

Question 155

```
# In helper.py: def square(n): return n*n
# In main.py:
# from helper import square
# if __name__ == "__main__": print(square(5))
```

Files, exceptions, JSON, and regular expressions

Question 156

```
from pathlib import Path
p=Path("notes.txt"); p.write_text("First line\nSecond line\n",encoding="utf-8"); print(p.read_text(
encoding="utf-8"))
```

Question 157

```
from pathlib import Path
p=Path(input("Filename: ")); text=input("Text to append: ")
with p.open("a",encoding="utf-8") as f: f.write(text+"\n")
```

Question 158

```
from pathlib import Path
text=Path("notes.txt").read_text(encoding="utf-8"); print("lines",len(text.splitlines()),"words",len(
text.split()),"characters",len(text))
```

Question 159

```
from pathlib import Path
src,dst=Path("source.txt"),Path("copy.txt")
try: dst.write_text(src.read_text(encoding="utf-8"),encoding="utf-8")
except FileNotFoundError: print("Source file not found")
```

Question 160

```
from pathlib import Path
word=input("Word: ")
for no,line in enumerate(Path("notes.txt").read_text(encoding="utf-8").splitlines(),1):
 if word in line: print(no,line)
```

Question 161

```
from pathlib import Path
p=Path("notes.txt"); text=p.read_text(encoding="utf-8"); old=input("Old word: "); new=input("New word:
"); Path("updated.txt").write_text(text.replace(old,new),encoding="utf-8")
```

Question 162

```
from pathlib import Path
folder=Path("data"); folder.mkdir(parents=True,exist_ok=True); (folder/"example.txt").write_text("Hello",
encoding="utf-8")
```

Question 163

```
from pathlib import Path
p=Path("old.txt"); new=Path("new.txt")
if p.is_file(): p.rename(new)
if new.is_file(): new.unlink()
```

Question 164

```
try:
 with open("notes.txt",encoding="utf-8") as f: print(f.read())
except FileNotFoundError: print("File does not exist.")
except PermissionError: print("Permission denied.")
```

Question 165

```
import json
from pathlib import Path
data={"name":"Asha","scores":[90,85]}; p=Path("data.json"); p.write_text(json.dumps(data,indent=2),
encoding="utf-8")
try: print(json.loads(p.read_text(encoding="utf-8")))
except json.JSONDecodeError: print("Invalid JSON")
```

Question 166

```
import json
from pathlib import Path
p=Path("contacts.json")
contacts=json.loads(p.read_text()) if p.exists() else {}
contacts["Asha"]="123456"; p.write_text(json.dumps(contacts,indent=2)); print(contacts)
```

Question 167

```
import csv
with open("data.csv",newline="",encoding="utf-8") as f:
 for row in csv.reader(f): print(row)
```

Question 168

```
import csv
with open("data.csv",newline="",encoding="utf-8") as f: rows=list(csv.DictReader(f))
values=[float(row["amount"]) for row in rows if row.get("amount")]
print("total",sum(values),"average",sum(values)/len(values) if values else 0)
```

Question 169

```
import re
text=input("Text: "); print(re.findall(r"[\w.+-]+@[\w.-]+\.[A-Za-z]{2,}",text))
```

Question 170

```
import re
text=input("Text: "); print(re.findall(r"-?\d+(?:\.\d+)?",text))
```

Question 171

```
import re
s=input("Date DD-MM-YYYY: "); match=re.fullmatch(r"\d{2}-\d{2}-\d{4}",s); print(bool(match))
# For a real calendar-date check, also parse with datetime.strptime.
```

Question 172

```
import logging
logging.basicConfig(filename="errors.log",level=logging.ERROR)
from pathlib import Path
try: print(Path(input("Filename: ")).read_text(encoding="utf-8"))
except (OSError,UnicodeError) as err: logging.exception("Could not read file"); print(f"Could not read
file: {err}")
```

Object-oriented programming

Question 173

```
class Student:
 def __init__(self,name,marks): self.name,self.marks=name,marks
 def display(self): print(f"{self.name}: {self.marks}")
Student("Asha",92).display()
```

Question 174

```
class Rectangle:
 def __init__(self,l,w): self.l,self.w=l,w
 def area(self): return self.l*self.w
 def perimeter(self): return 2*(self.l+self.w)
r=Rectangle(3,4); print(r.area(),r.perimeter())
```

Question 175

```
class BankAccount:
 def __init__(self,balance=0): self.balance=balance
 def deposit(self,amount):
  if amount<=0: raise ValueError("Amount must be positive")
  self.balance+=amount
 def withdraw(self,amount):
  if amount<=0 or amount>self.balance: raise ValueError("Invalid withdrawal")
  self.balance-=amount
```

Question 176

```
class User:
 def __init__(self,name,role="member"): self.name,self.role=name,role
print(User("Asha").role,User("Ravi","admin").role)
```

Question 177

```
class Example:
 category="demo"
 def __init__(self,value): self.value=value
a=Example(1); b=Example(2); print(a.value,b.value,a.category)
```

Question 178

```
class Student:
 def __init__(self,name,marks): self.name,self.marks=name,marks
 def __str__(self): return f"{self.name} (marks: {self.marks})"
print(Student("Asha",90))
```

Question 179

```
class Temperature:
 def __init__(self,celsius=0): self.set_celsius(celsius)
 def set_celsius(self,value):
  if value < -273.15: raise ValueError("Below absolute zero")
  self.celsius=value
```

Question 180

```
class BankAccount:
 def __init__(self,balance=0): self.balance=balance
class SavingsAccount(BankAccount):
 def __init__(self,balance=0,rate=0.02): super().__init__(balance); self.rate=rate
```

Question 181

```
class Animal:
 def speak(self): return "sound"
class Dog(Animal):
 def speak(self): return "woof"
print(Animal().speak(),Dog().speak())
```

Question 182

```
class Person:
 def __init__(self,name): self.name=name
class Student(Person):
 def __init__(self,name,student_id): super().__init__(name); self.student_id=student_id
print(Student("Asha","S01").name)
```

Question 183

```
from abc import ABC,abstractmethod
class Shape(ABC):
 @abstractmethod
 def area(self): pass
class Square(Shape):
 def __init__(self,side): self.side=side
 def area(self): return self.side**2
print(Square(4).area())
```

Question 184

```
class Dog:
 def speak(self): return "woof"
class Cat:
 def speak(self): return "meow"
for animal in [Dog(),Cat()]: print(animal.speak())
```

Question 185

```
class Person:
 def __init__(self,age): self.age=age
 @property
 def age(self): return self._age
 @age.setter
 def age(self,value):
  if value<0: raise ValueError("Age cannot be negative")
  self._age=value
```

Question 186

```
class Basket:
 def __init__(self,items): self.items=items
 def __len__(self): return len(self.items)
 def __eq__(self,other): return isinstance(other,Basket) and self.items==other.items
print(len(Basket([1,2])))
```

Question 187

```
class Book:
 def __init__(self,title): self.title=title; self.available=True
 def borrow(self):
  if not self.available: return False
  self.available=False; return True
 def return_book(self): self.available=True
```

Question 188

```
class Product:
 def __init__(self,name,stock=0): self.name=name; self.stock=stock
 def add(self,n):
  if n>0: self.stock+=n
 def sell(self,n):
  if 0<n<=self.stock: self.stock-=n; return True
  return False
```

Question 189

```
class Engine:
 def start(self): return "engine started"
class Car:
 def __init__(self): self.engine=Engine()
car=Car(); print(car.engine.start())
```

Question 190

```
class ExpenseTracker:
 def __init__(self): self.expenses=[]
 def add(self,description,amount):
  if amount<0: raise ValueError("Amount must be non-negative")
  self.expenses.append((description,amount))
 def total(self): return sum(amount for _,amount in self.expenses)
t=ExpenseTracker(); t.add("Food",12); print(t.total())
```

Modules, packages, and integration projects

Question 191

```
import math
r=float(input("Radius: ")); print("area",math.pi*r*r,"sqrt",math.sqrt(r) if r>=0 else "undefined")
```

Question 192

```
import random
secret=random.randint(1,100); attempts=0
while True:
 guess=int(input("Guess 1-100: ")); attempts+=1
 if guess==secret: print("Correct in",attempts,"attempts"); break
 print("Too low" if guess<secret else "Too high")
```

Question 193

```
from datetime import date
start=date.fromisoformat(input("Start date YYYY-MM-DD: ")); end=date.fromisoformat(input("End date YYYY-
MM-DD: ")); print(abs((end-start).days),"days apart")
```

Question 194

```
import time
start=time.perf_counter()
result=sum(range(1_000_000))
elapsed=time.perf_counter()-start
print(result, f"elapsed: {elapsed:.6f}s")
```

Question 195

```
from pathlib import Path
folder=Path(input("Directory: "))
for p in folder.iterdir():
 if p.is_file() and p.suffix.lower()==".txt": print(p.name)
```

Question 196

```
import shutil
from pathlib import Path
src=Path("source.txt"); dst=Path("backup/source.txt")
try: dst.parent.mkdir(parents=True,exist_ok=True); shutil.copy2(src,dst)
except OSError as e: print("Backup failed:",e)
```

Question 197

```
import statistics
values=list(map(float,input("Values: ").split()))
if values: print("mean",statistics.mean(values),"median",statistics.median(values))
else: print("Please enter at least one value")
```

Question 198

```
import sys
if len(sys.argv)!=2: print(f"Usage: {sys.argv[0]} VALUE"); raise SystemExit(2)
print("Value:",sys.argv[1])
```

Question 199

```
# helpers.py: def double(n): return n*2
# main.py:
# from helpers import double
# print(double(21))
```

Question 200

```
import csv,json,statistics
from pathlib import Path
source=Path("transactions.csv"); report=Path("report.json")
with source.open(newline="",encoding="utf-8") as f: rows=list(csv.DictReader(f))
amounts=[float(r["amount"]) for r in rows if r.get("amount")]
summary={"count":len(amounts),"total":sum(amounts),"mean":statistics.mean(amounts) if amounts else 0,
"median":statistics.median(amounts) if amounts else 0}
report.write_text(json.dumps(summary,indent=2),encoding="utf-8"); print(summary)
```